Quiz 19 Solution **Roll number:** _____

1. **What happens if you forget to free dynamically allocated memory in C++?**
   a) The program will crash immediately.
   b) The memory will automatically be released when the function returns.
   c) A memory leak occurs.
   d) The program will run slower.

2. **Which of the following is true regarding dynamic arrays?**
   a) Their size is fixed at compile time.
   b) They can only hold primitive data types.
   c) They allow for flexible sizing at runtime.
   d) They do not require memory management.

3. **What will happen if you use `delete` on a pointer that was allocated with `new[]`?**
   a) It will cause a memory leak.
   b) It will delete the entire array correctly.
   c) It will lead to undefined behavior.
   d) It will free only the first element of the array.

Write a C++ code that initializes a dynamic memory array of size 10. The program should prompt the user to input values to store in this array. After the values have been entered, display sum of all numbers and ensure that the allocated memory for the array is properly deallocated.

```
#include <iostream>
using namespace std;
int main() {
    int sum = 0;
    int * arr = new int [10];
    for (int i=0; i<10; i++) {
        cout << "Enter " << i+1 << " number: ";
        cin >> arr [i];
    }
    for (int i=0; i<10; i++) {
        sum += arr [i];
    }
    cout << " Sum is : " << sum << endl;
    delete [] arr;
    arr = nullptr;
    return 0;
}
```

⇒ 1 mark

⇒ 1 marks

⇒ 1.5 mark

⇒ 1.5 marks

1 mark ⇐

⇒ 1 mark

Marks will be only given if you do operations on dma array not on static array

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